

Introduction to Coding theory

Ji, Yonghyeon

April 4, 2026

We cover the following topics in this note.

- TBA.

Contents

1	Error-Correcting Codes	2
1.1	Linear Codes	4
A	Exercises	6

1 Error-Correcting Codes

The fundamental problem of coding theory is to transmit information reliably across a noisy channel. A sender begins with a message, usually represented as a vector

$$\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_2^k,$$

and applies an *encoder*

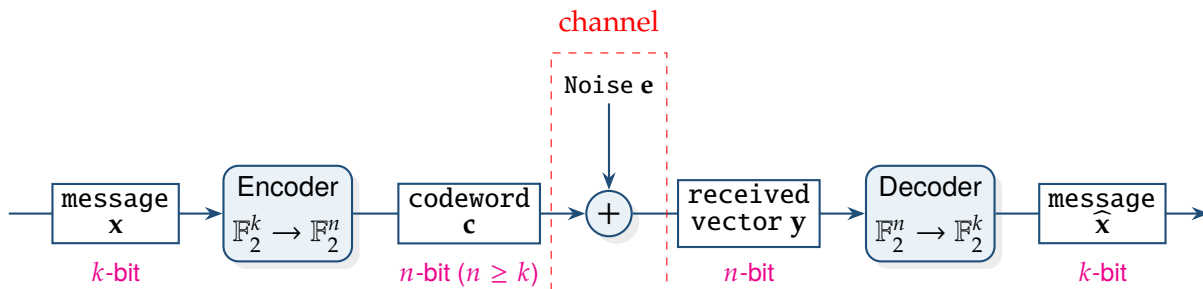
$$\text{Encode} : \mathbb{F}_2^k \longrightarrow \mathbb{F}_2^n, \quad \mathbf{x} \longmapsto \mathbf{c},$$

where $n \geq k$. The vector $\mathbf{c}$ is called the *codeword*.

During transmission, the codeword passes through a noisy channel. The effect of the channel is modeled by adding an error vector $\mathbf{e} \in \mathbb{F}_2^n$ to the transmitted codeword, so that the receiver obtains

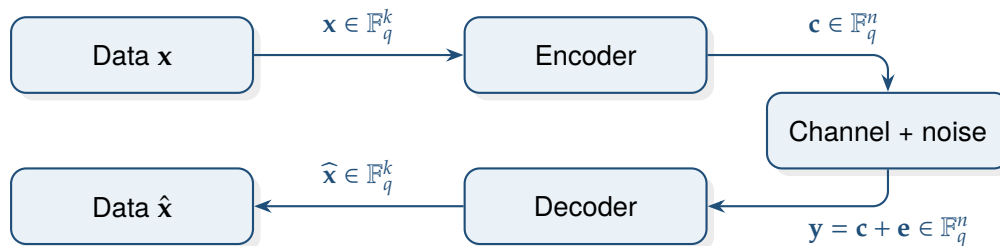
$$\mathbf{y} = \mathbf{c} \oplus \mathbf{e} \in \mathbb{F}_2^n.$$

The vector $\mathbf{y}$ is called the *received vector*. The purpose of the decoder is to recover the original message, or equivalently the transmitted codeword, from the corrupted vector $\mathbf{y}$.



A generic communication model has the form

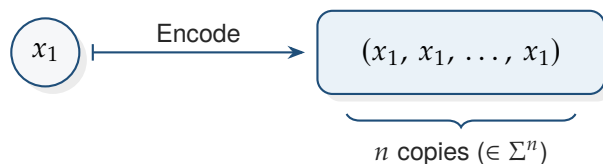
$$\mathbf{x} = (x_1, \dots, x_k) \xrightarrow{\text{encoder}} \underbrace{\mathbf{c} = (c_1, \dots, c_n)}_{\text{codeword } (n \geq k)} \xrightarrow{\text{noise } \mathbf{e}} \mathbf{y} = \mathbf{c} + \mathbf{e} \xrightarrow{\text{decoder}} \hat{\mathbf{x}} = (\hat{x}_1, \dots, \hat{x}_k).$$



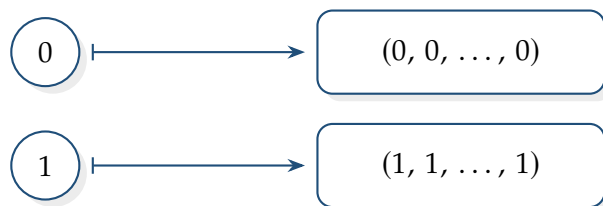
Example 1 (Repetition code). Let Σ be an alphabet. An *encoding* is a map

$$\begin{aligned} \text{Encode} : \quad \Sigma^k &\longrightarrow \Sigma^n \\ (x_1, \dots, x_k) &\longmapsto (c_1, \dots, c_n) \end{aligned}$$

with $n \geq k$.



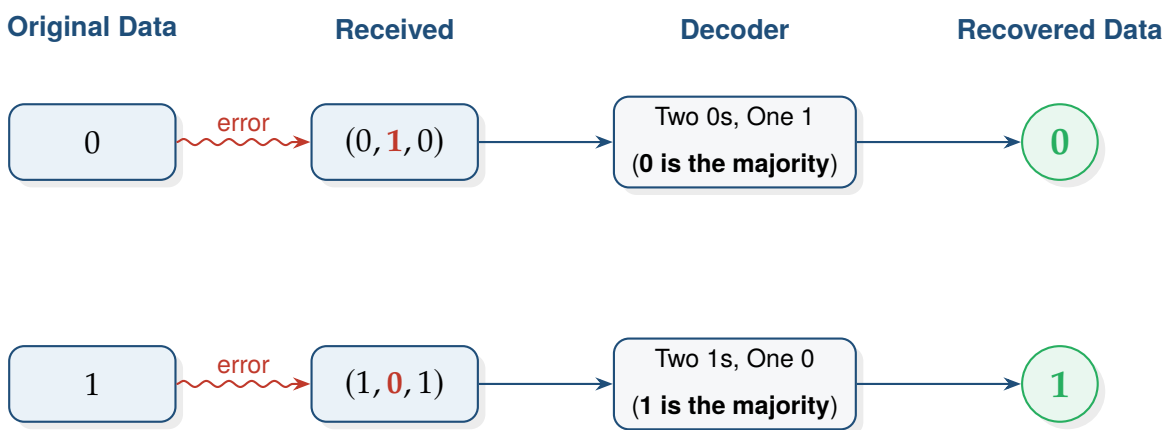
When $k = 1$, a very simple encoding repeats the same symbol



This is called the *repetition code*. If one symbol is changed during transmission, we can still recover the original message by *majority vote*. For instance,

$$(0, 0, 0) \rightsquigarrow (0, 1, 0), \quad (1, 1, 1) \rightsquigarrow (1, 0, 1).$$

Since 0 appears most often in $(0, 1, 0)$, we decode it as 0, and since 1 appears most often in $(1, 0, 1)$, we decode it as 1.



1.1 Linear Codes

Linear Encoding

Definition 1. A **linear encoding** over $\mathbb{F}_q$ is a linear map

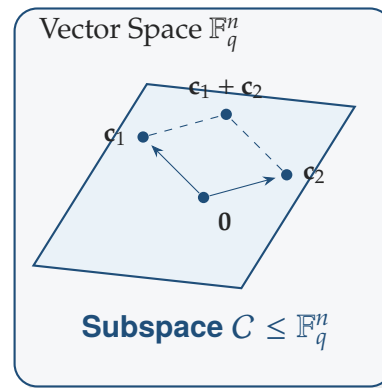
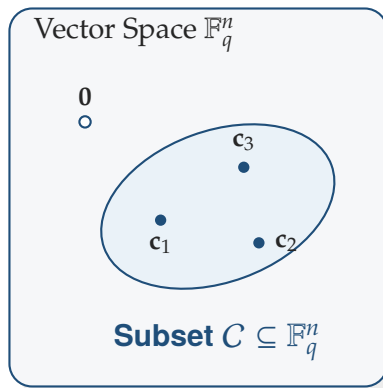
$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \quad \mathbf{x} \longmapsto \mathbf{c}.$$

Linear Code

Definition 2. A **code of length** $n \in \mathbb{Z}_{>0}$ **over** $\mathbb{F}_q$ is a subset

$$C \subseteq \mathbb{F}_q^n.$$

The elements of C are called **codewords**. If $C \leq \mathbb{F}_q^n$, then C is called a **linear code**.



Remark 1. If $\dim_{\mathbb{F}_q}(C) = k$, then C is called a $[n, k]_q$ -**code over** $\mathbb{F}_q$.

Proposition 1. If C is an $[n, k]_q$ -code, then

$$|C| = q^k.$$

Proof. Choose a basis $\mathbf{b}_1, \dots, \mathbf{b}_k$ of C . Every codeword can be written uniquely as

$$x_1 \mathbf{b}_1 + \dots + x_k \mathbf{b}_k, \quad x_i \in \mathbb{F}_q.$$

Each x_i has q choices, and there are k independent coefficients. Therefore the number of codewords is q^k . $\square$

A Exercises